

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
#Setup
library(tidyverse)
library(lubridate)
library(trend)
library(zoo)
library(Kendall)
library(ggplot2)
library(here)
here()

#Set theme
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "top")
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only)

allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named `GaringerOzone` of 3589 observation and 20 variables.

```
#1 Importing datasets
data1 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv",
  stringsAsFactors = TRUE)
data2 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv",
  stringsAsFactors = TRUE)
data3 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv",
  stringsAsFactors = TRUE)
data4 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv",
  stringsAsFactors = TRUE)
data5 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv",
  stringsAsFactors = TRUE)
data6 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv",
  stringsAsFactors = TRUE)
data7 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv",
  stringsAsFactors = TRUE)
data8 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv",
  stringsAsFactors = TRUE)
data9 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv",
  stringsAsFactors = TRUE)
data10 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv",
  stringsAsFactors = TRUE)

#2 Combining datasets
GaringerOzone <- bind_rows(data1, data2, data3, data4, data5, data6, data7, data8, data9, data10)
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns `Date`, `Daily.Max.8.hour.Ozone.Concentration`, and `DAILY_AQI_VALUE`.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame `Days`. Rename the column name in `Days` to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame `GaringerOzone`.

```
#3 Dates
GaringerOzone$Date <- mdy(GaringerOzone$Date)

#4 Wrangling data
garinger.sort <- GaringerOzone %>%
  select(c("Date", "Daily.Max.8.hour.Ozone.Concentration", "DAILY_AQI_VALUE"))

#5 Dates 2
Days <- as.data.frame(seq(from = as.Date("2010-01-01"),
  to = as.Date("2019-12-31"),
  by = "day"))
```

```
colnames(Days) <- "Date"

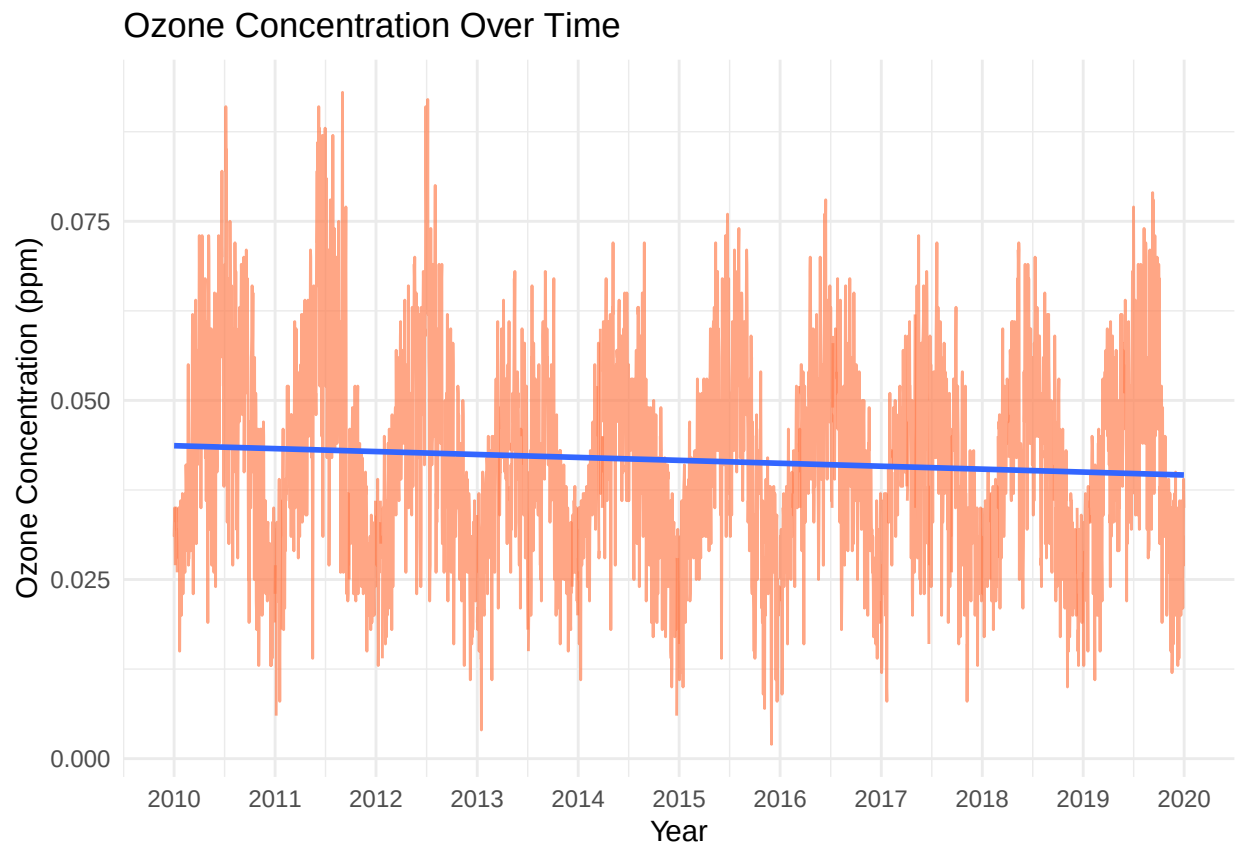
#6 Combining
GaringerOzone <-
  left_join(Days, garinger.sort, by = "Date")
```

Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7 Graphing
O3_conc <-
  ggplot(GaringerOzone, aes(y = Daily.Max.8.hour.Ozone.Concentration, x = Date)) +
  geom_line(color = "coral", alpha = 0.7) +
  labs(y = "Ozone Concentration (ppm)",
       x = "Year",
       title = "Ozone Concentration Over Time") +
  scale_x_date(date_labels = "%Y", date_breaks = "1 year") +
  geom_smooth(method = lm, se = FALSE) +
  theme_minimal()

print(O3_conc)
```



Answer: The plot suggests a slight downward trend in ozone concentration over time, as indicated by the blue linear regression line. While ozone levels exhibit seasonal fluctuations, with peaks and troughs occurring annually, the overall trend appears to be declining.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8 Filling missing data
GaringerOzone_clean <- GaringerOzone %>%
  mutate(conc_clean = na.approx(Daily.Max.8.hour.Ozone.Concentration))
```

Answer: We use linear interpolation to fill missing ozone data because it maintains a smooth and realistic transition between observed values. Piecewise constant interpolation would hold the last recorded value, which is unrealistic for continuously varying ozone levels. Spline interpolation, while smooth, could introduce artificial fluctuations. Linear interpolation is the best choice as it preserves the natural trend of ozone concentration over time without overfitting.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9 Grouping
GaringerOzone.monthly <- GaringerOzone_clean %>%
  mutate(
    Year = year(Date),
    Month = month(Date)
  ) %>%
  group_by(Year, Month) %>%
  summarise(Mean_Ozone = mean(conc_clean, na.rm = TRUE), .groups = "drop")

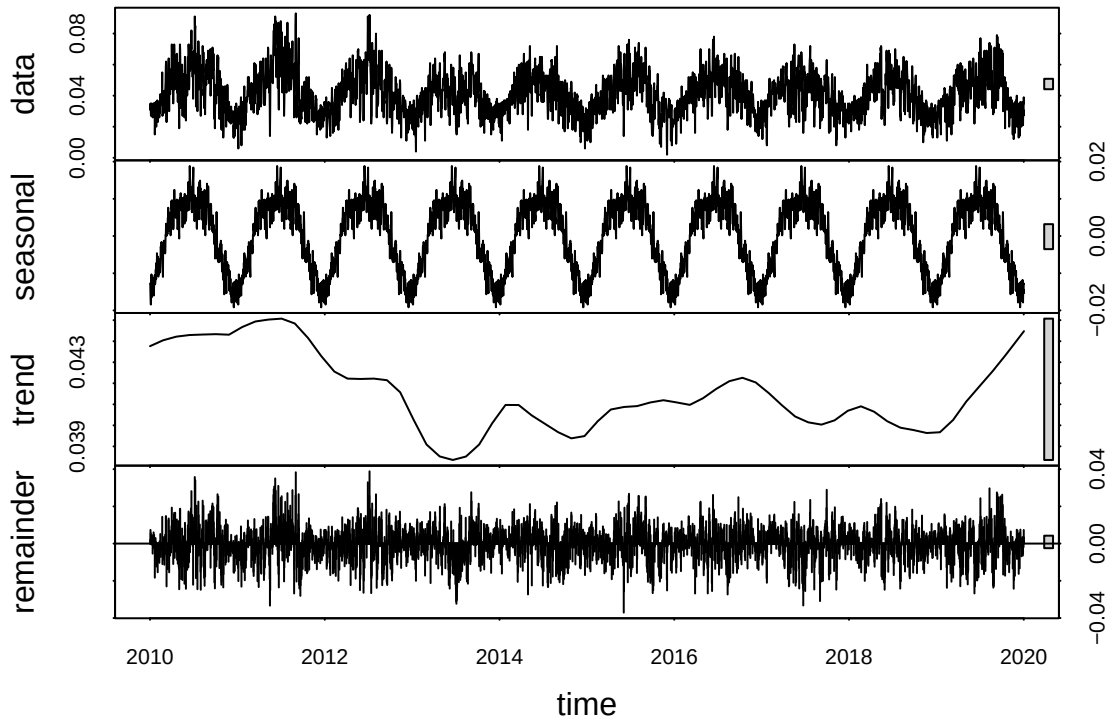
GaringerOzone.monthly <- GaringerOzone.monthly %>%
  mutate(Date = as.Date(paste(Year, Month, "01", sep = "-")))
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

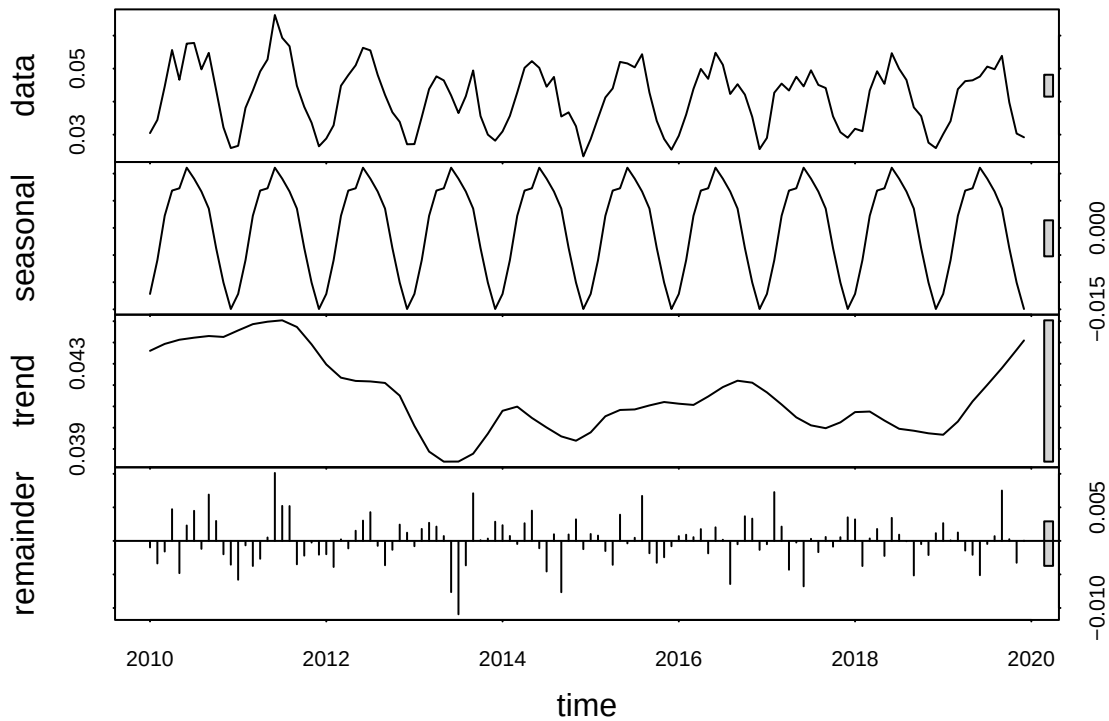
```
#10 Generating time series
GaringerOzone.daily.ts <- ts(GaringerOzone_clean$conc_clean,
  start = c(2010,1),
  frequency = 365)
GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$Mean_Ozone,
  start = c(2010,1),
  frequency = 12)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11 Decomposing
daily_decomp <- stl(GaringerOzone.daily.ts,s.window = "periodic")
plot(daily_decomp)
```



```
monthly_decomp <- stl(GaringerOzone.monthly.ts,s.window = "periodic")
plot(monthly_decomp)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

#12 Trend analysis

```
monthly_data_trend <- Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)
monthly_data_trend
```

```
## tau = -0.143, 2-sided pvalue =0.046724
```

```
summary(monthly_data_trend)
```

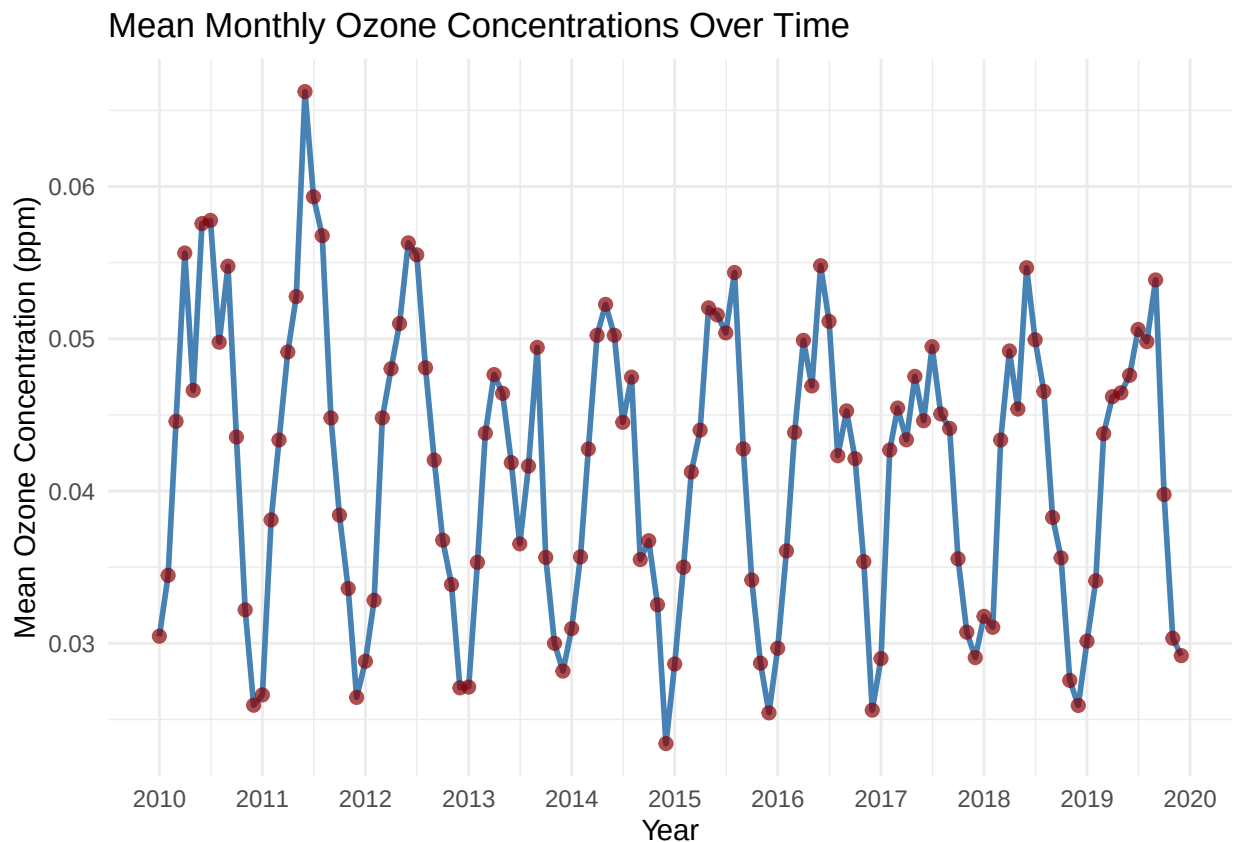
```
## Score = -77 , Var(Score) = 1499
## denominator = 539.4972
## tau = -0.143, 2-sided pvalue =0.046724
```

Answer: The Seasonal Mann-Kendall test is ideal for analyzing ozone trends because it accounts for seasonality, which is evident in the decomposition plot.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

13 Graphing

```
meanO3 <-  
  ggplot(GaringerOzone.monthly, aes(x = Date, y = Mean_Ozone)) +  
    geom_line(color = "steelblue", linewidth = 1) +  
    labs(x = "Year",  
         y = "Mean Ozone Concentration (ppm)",  
         title = "Mean Monthly Ozone Concentrations Over Time") +  
    scale_x_date(date_labels = "%Y", date_breaks = "1 year") +  
    geom_point(color = "darkred", size = 2, alpha = 0.7) +  
    theme_minimal()  
  
print(meanO3)
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: The time series plot shows strong seasonal variation in ozone concentrations, with recurring peaks and troughs each year. However, the Seasonal Mann-Kendall test detects a statistically significant downward trend in ozone levels over the 2010s ($\tau = -0.143$, $p = 0.047$), indicating a gradual decline in concentrations. This suggests that ozone levels at this station have been decreasing over time, possibly due to regulatory policies, emission reductions, or atmospheric changes.

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.

16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15 Extracting data
monthly_decomp_component <-
  monthly_decomp$time.series[, "seasonal"]

GaringerOzone.monthly.noseason <- GaringerOzone.monthly.ts - monthly_decomp_component

#16 Mann Kendall test
monthly_data_trend_noseason <- Kendall::MannKendall(GaringerOzone.monthly.noseason)
monthly_data_trend_noseason
```

```
## tau = -0.165, 2-sided pvalue =0.0075402
```

```
summary(monthly_data_trend_noseason)
```

```
## Score = -1179 , Var(Score) = 194365.7
## denominator = 7139.5
## tau = -0.165, 2-sided pvalue =0.0075402
```

Answer: The Seasonal Mann-Kendall test on the original series found a moderate downward trend ($\tau = -0.143$, $p = 0.0467$). After removing seasonality, the trend became stronger ($\tau = -0.165$, $p = 0.0075$), confirming a more consistent long-term decline in ozone levels. This suggests that seasonal fluctuations masked part of the trend, and the deseasonalized data provides clearer evidence of decreasing ozone concentrations.